

1. (1 point) Select all that applies. The equipollence relation is
 A. an equivalence relation B. antisymmetric C. transitive D. a partial order
2. (1 point) Let A' , B' and C' be midpoints of the sides of a triangle ABC . We have

$$\overrightarrow{OA'} + \overrightarrow{OB'} + \overrightarrow{OC'} = c \cdot (\overrightarrow{OA} + \overrightarrow{OB} + \overrightarrow{OC}).$$

for which value c ? _____

3. (1 point) Let $ABCD$ be a tetrahedron. Which of the following equalities are true?

- A. $\overrightarrow{AD} = \overrightarrow{AB} + \overrightarrow{BC} + \overrightarrow{CD}$
 B. $\overrightarrow{AC} = \overrightarrow{AD} + \overrightarrow{BC} + \overrightarrow{DB}$
 C. $\overrightarrow{AB} = \overrightarrow{AD} + \overrightarrow{BC} + \overrightarrow{DB}$
 D. $0 = \overrightarrow{AB} + \overrightarrow{CD} + \overrightarrow{BC} + \overrightarrow{DA}$

4. (1 point) Let $ABCD$ be a tetrahedron. For which vector $\mathbf{v}$ do we have $\overrightarrow{AD} + \overrightarrow{BC} = \overrightarrow{BD} + \mathbf{v}$?

- A. $\overrightarrow{CB}$ B. $\overrightarrow{AB}$ C. $\overrightarrow{AC}$ D. $\overrightarrow{CD}$

5. (1 point) Let $SABCD$ be a pyramid with apex S and base the parallelogram $ABCD$. Let O be the center of the parallelogram. We have

$$\overrightarrow{SA} + \overrightarrow{SB} + \overrightarrow{SC} + \overrightarrow{SD} = c \cdot \overrightarrow{SO}$$

for which value c ? _____

6. (1 point) Let $a \in \mathbb{R}$. With respect to the basis $\mathcal{B} = (\mathbf{i}, \mathbf{j}, \mathbf{k})$ consider the vectors $\mathbf{u} = \mathbf{i} + \mathbf{j}$, $\mathbf{v} = \mathbf{j} + a\mathbf{k}$ and $\mathbf{w} = \mathbf{i} + \mathbf{k}$. The values a for which $(\mathbf{u}, \mathbf{v}, \mathbf{w})$ is a basis with the same orientation as $\mathcal{B}$ are _____

7. (1 point) With respect to a right oriented frame $\mathcal{K} = (O, \mathcal{B})$ of $\mathbb{E}^3$, consider the vertices $A(1, 2, 0)$, $B(1, 2, 1)$, $C(0, 3, 2)$, $D(1, 1, 1)$ of a tetrahedron. Let $\mathcal{K}'$ be the coordinate system with origin in B and basis $\mathcal{B}' = (\overrightarrow{AB}, \overrightarrow{AC}, \overrightarrow{AD})$. Which of the following are true?

- A. The coordinates of A with respect to $\mathcal{K}'$ are $(-1, 0, 0)$
 B. $\mathcal{K}$ and $\mathcal{K}'$ have the same orientation
 C. The components of $\overrightarrow{AB}$ with respect to $\mathcal{B}$ are $(0, 0, 1)$
 D. The components of $\overrightarrow{BC}$ with respect to $\mathcal{B}'$ are $(-1, 1, 0)$

Setup 1: Let $\mathcal{K} = (O, \mathcal{B})$ be an frame of $\mathbb{E}^2$. Let $\mathcal{K}' = (O', \mathcal{B}')$ be a second frame with $\mathcal{B}' = (\mathbf{i}', \mathbf{j}')$ and

$$[O']_{\mathcal{K}} = \begin{bmatrix} -2 \\ -2 \end{bmatrix}, \quad [\mathbf{i}']_{\mathcal{B}} = \begin{bmatrix} 3 \\ -1 \end{bmatrix}, \quad [\mathbf{j}']_{\mathcal{B}} = \begin{bmatrix} 0 \\ -1 \end{bmatrix}.$$

Consider the points

$$[A]_{\mathcal{K}} = \begin{bmatrix} -5 \\ 2 \end{bmatrix}, \quad [B]_{\mathcal{K}} = \begin{bmatrix} -2 \\ 5 \end{bmatrix}, \quad [C]_{\mathcal{K}} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

8. (1 point) Consider Setup 1 above. The coordinates of B with respect to the frame $\mathcal{K}'$ are _____

9. (1 point) Consider Setup 1 above. Which of the following bases have the same orientation as $\mathcal{B}$?

A. $(\overrightarrow{CB}, \overrightarrow{CA})$ B. $(\overrightarrow{BA}, \overrightarrow{BC})$ C. $(\overrightarrow{AB}, \overrightarrow{AC})$ D. $(\overrightarrow{AC}, \overrightarrow{AB})$

Setup 2: Let $\mathcal{K} = (O, \mathcal{B})$ be a right-oriented orthonormal frame of $\mathbb{E}^2$. Let $\mathcal{K}' = (O', \mathcal{B}')$ be a second frame with $\mathcal{B}' = (\mathbf{i}', \mathbf{j}', \mathbf{k}')$ and

$$[O']_{\mathcal{K}} = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}, \quad [\mathbf{i}']_{\mathcal{B}} = \begin{bmatrix} 2 \\ 0 \\ -1 \end{bmatrix}, \quad [\mathbf{j}']_{\mathcal{B}} = \begin{bmatrix} 0 \\ -1 \\ 0 \end{bmatrix}, \quad [\mathbf{k}']_{\mathcal{B}} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}.$$

Consider the points

$$[A]_{\mathcal{K}} = \begin{bmatrix} 1 \\ -1 \\ -1 \end{bmatrix}, \quad [B]_{\mathcal{K}} = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}, \quad [C]_{\mathcal{K}} = \begin{bmatrix} -1 \\ 2 \\ 0 \end{bmatrix}, \quad [D]_{\mathcal{K}} = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}.$$

10. (1 point) Consider Setup 2 above. The coordinates of D with respect to the frame $\mathcal{K}'$ are _____
11. (1 point) Consider Setup 2 above. Which of the following bases have the same orientation as $\mathcal{B}$?
- A. $(\overrightarrow{AB}, \overrightarrow{BC}, \overrightarrow{BD})$ B. $(\overrightarrow{BA}, \overrightarrow{BC}, \overrightarrow{BD})$ C. $(\overrightarrow{AB}, \overrightarrow{AC}, \overrightarrow{AD})$ D. $(\overrightarrow{AC}, \overrightarrow{AB}, \overrightarrow{AD})$

Answers:

1. A, C
2. $c = 1$
3. A, B, D
4. C
5. $c = 4$
6. $a > -1$
7. A, B, C, D
8. $(0, -7)$
9. A, B, D
10. $(1, 0, 1)$
11. A, C